Critical Design Review

Mars Hard Lander Electrical System

WATTS UP

Meet the team!

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Vision for Expo

🔋 Battery / ☀️ Solar

- Solar panel charges the battery during the demo using light source

📡 RF / Data

- Data is transmitted live over RF during EXPO

🧠 Mainboard

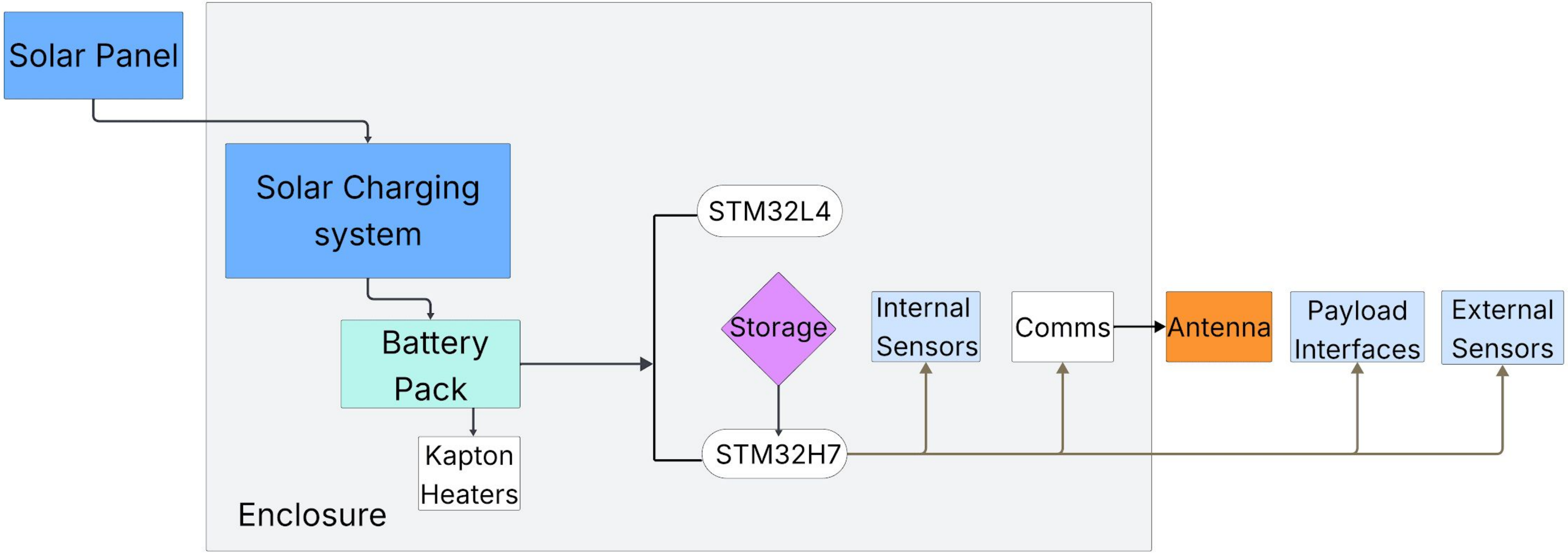
- Single mainboard manages solar charging, sensors, RF communication, and payloads

📦Enclosure Design

- Enclosure contains the battery, mainboard, sensors, and payload attachments

Final Design

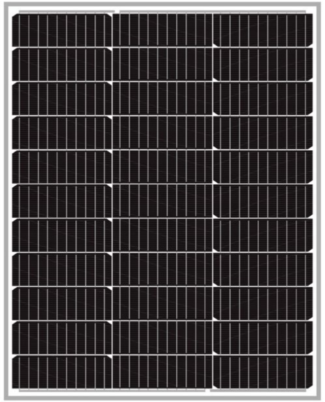
Functional Block Diagram



Power Hardware

Custom Battery:

- 1S9P pack
    - 3.7v nominal, 31.5 Ah, 116 Wh

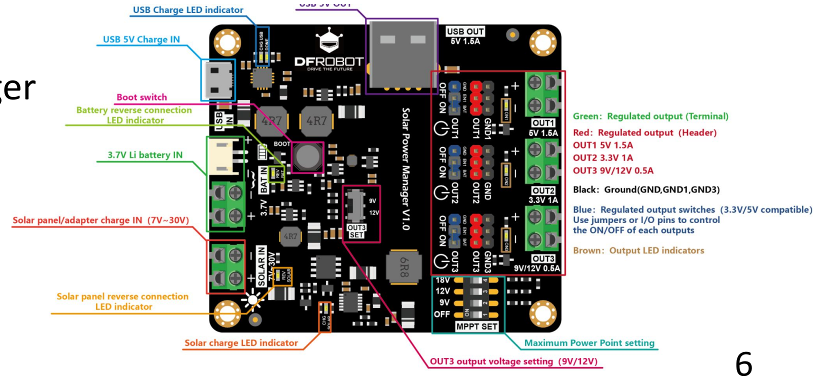


Solar Panel

- 100 W, I_mppt 5.32A, V_mppt = 18.81V
- I_SC = 5.6A, V_OC = 22.2V (~27V at -60°F)

Solar Manager:

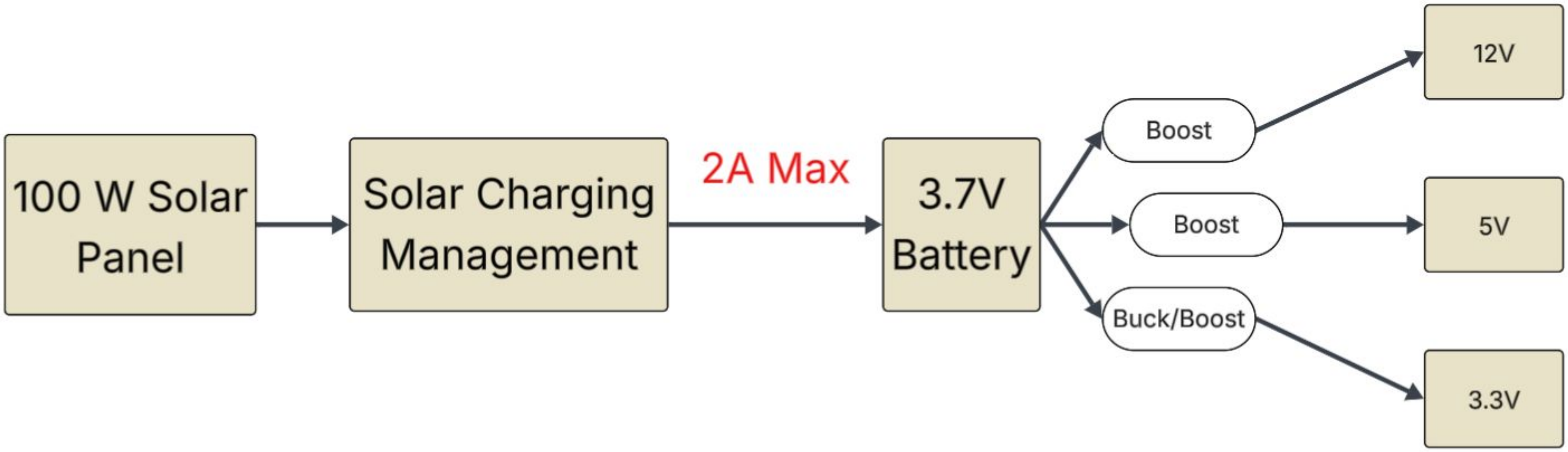
- Custom battery charger based of DFR Solar Power Manager
- Automatic switching between solar panel and battery



Kapton Heaters:

- Up to 12V at 0.4A → 4.8W

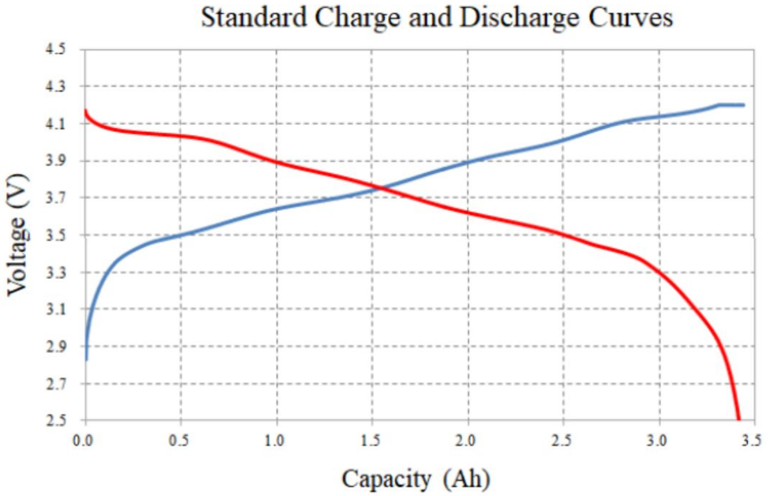
Power Structure

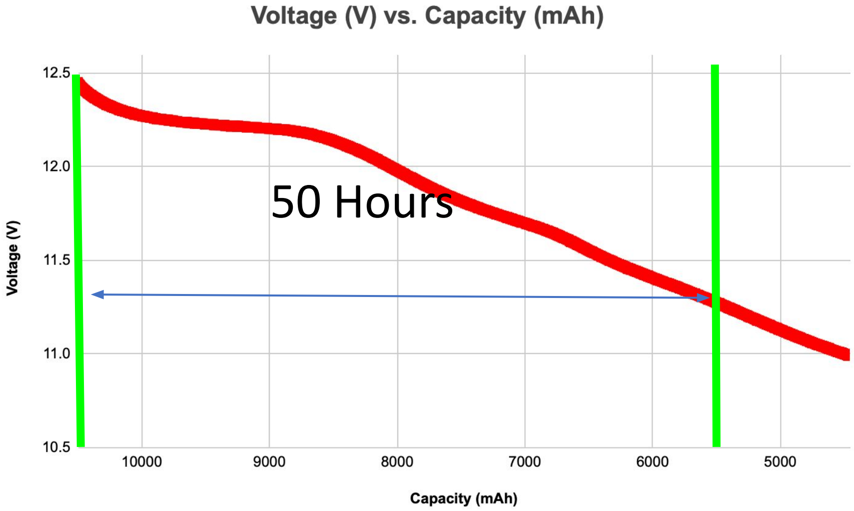


Battery Tests

Battery Discharge Test #1: (Performed with 12v Battery)

- Test performed at room temperature
- Replicated discharge curves from battery datasheet
- Passed MVP goal of 48.9 hours (Actual: 111 hours and 52 minutes)

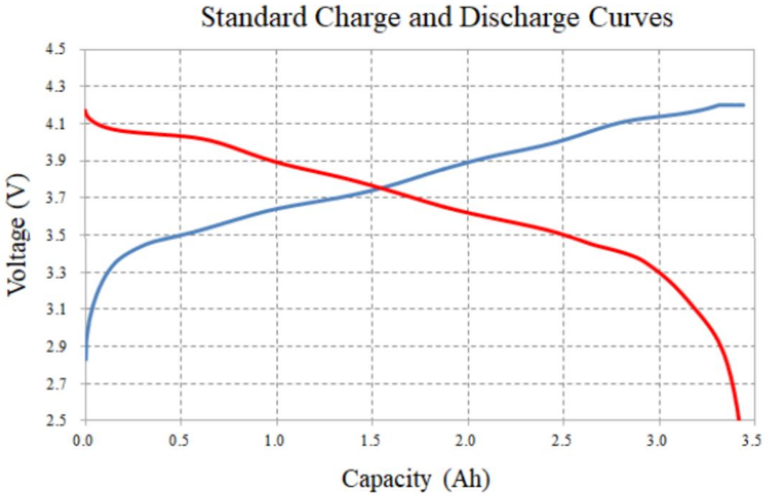


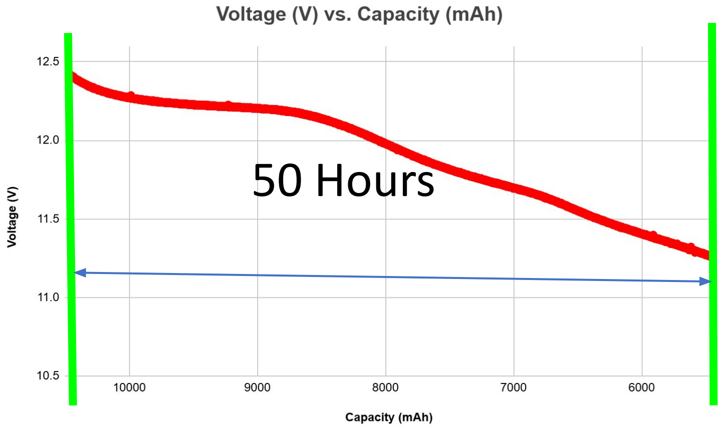


Battery Tests

Battery Discharge Test #2: (Performed with 12v Battery)

- Test performed at -40°F
- Passed MVP goal of 48.9 hours (Actual: 50 hours)





Future Tests & Risks

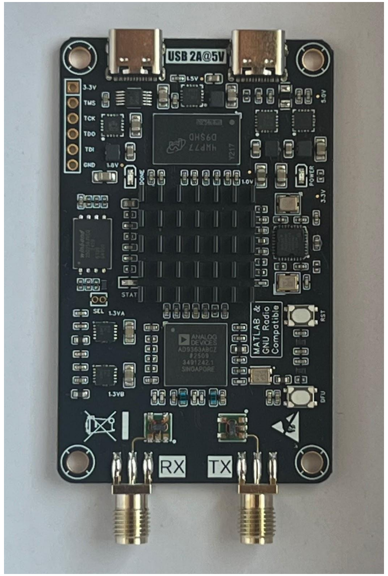
Future Tests:

1. Test solar panel charging board
2. Test battery heaters and battery discharge curves
3. Characterize loads and obtain nominal current draw

Power Subsystem Risks

1. Heaters draw too much current - **High**
2. Solar panel/battery not keeping load alive - **Med**

Radio



PlutoSDR Nano

- PlutoSDR Nano
  - ✔ Main communications board
    - 390-450Mhz
    - PSK Modulation
  - ✔ Interfaces with mainboard to transmit sensor data and interpret remote commands
- Post-Modulation Circuit
  - ✔ Implemented on Mainboard
  - ✔ External amplification and filtering
    - Up to 30dB TX output

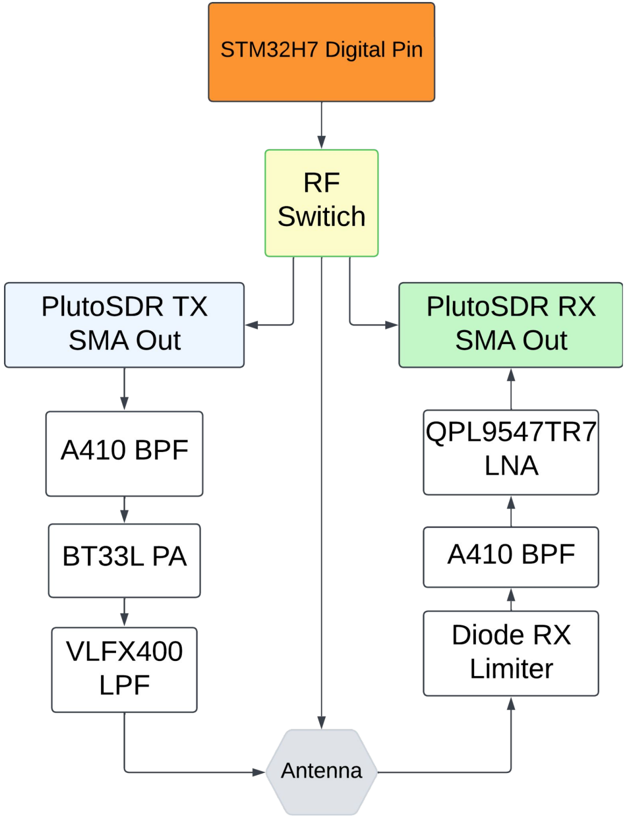
Radio Custom PCB

TX Channel

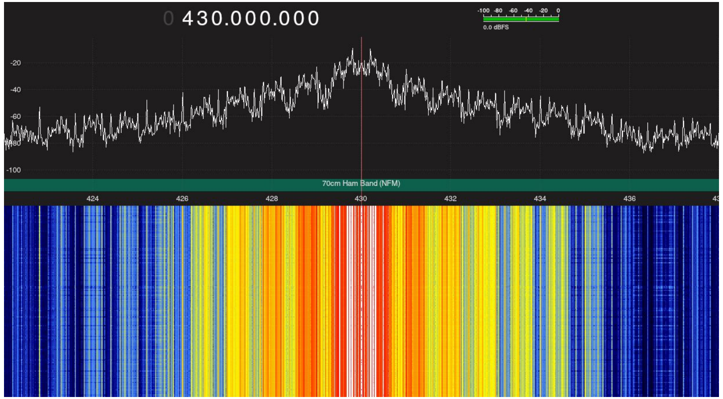
- Amplified to 30db
- Double filtering(Pre/Post) PA
- Decoupler protection for PA

RX Channel

- Protected from TX leakage via T/R switch
- Low-noise amplification
- Gain controlled digitally in Pluto SDR



Radio Testing



- Completed Tests
  - ✔ Basic transmission test in loopback configuration
    - Receive BPSK modulated waveform and demodulate back into bits
- Future Tests
  - ✔ PlutoSDR interfacing with mainboard
  - ✔ Native packetization on PlutoSDR board
  - ✔ Native receive testing on PlutoSDR board

Radio Software

- Application layer
  - ✔ Command Interpretation
  - ✔ Interfacing with mainboard
- Network layer
  - ✔ Packet creation/interpretation
- Data Link Layer
  - ✔ Error detection
  - ✔ I/Q Encoding/Decoding
- Physical Layer
  - ✔ Filtering
  - ✔ Interpolation/Decimation

Radio Hardware

Components:	TX channel	RX channel
Filters	A410 BPF pre PA LPF post PA	A410 BPF pre LNA
Amplifiers	BT33L PA (22 dB gain)	QPL9547TR7
Protection	ADL Decoupler	RX Limiter, PE42522B-X
Antenna	UHF 400-450 mHz	UHF 400-450 mHz

Radio Risks

- Full Protocol implementation - **High**
- Power delivery - **Med**
- Signal Integrity - **Med**
- Transmission errors (Data integrity) - **Low**
- Amplification - **Low**

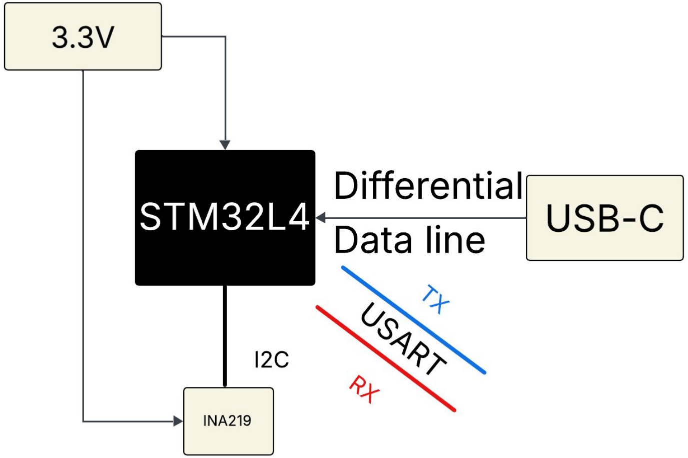
Low Power uC Subsystem

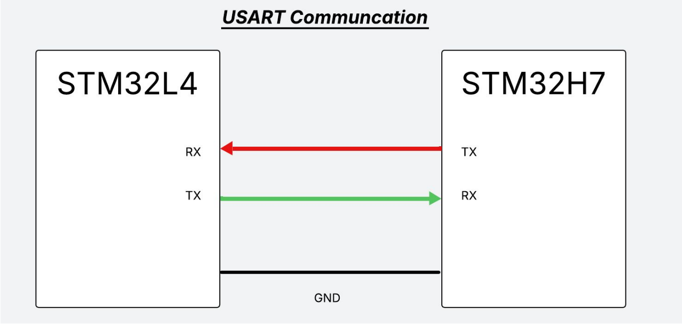
STM32L4 - Low Power Supervisor

- Always-on, ultra-low-power operation
- System power supervision and sequencing
- Controls STM32H7 power on/off
- Monitors battery and solar status
- Manages sleep / wake duty cycling

So What ?

- Preserves battery power by only waking up the STM32H7 (High Power) when data needs to be collected, and then sends power off command after that.

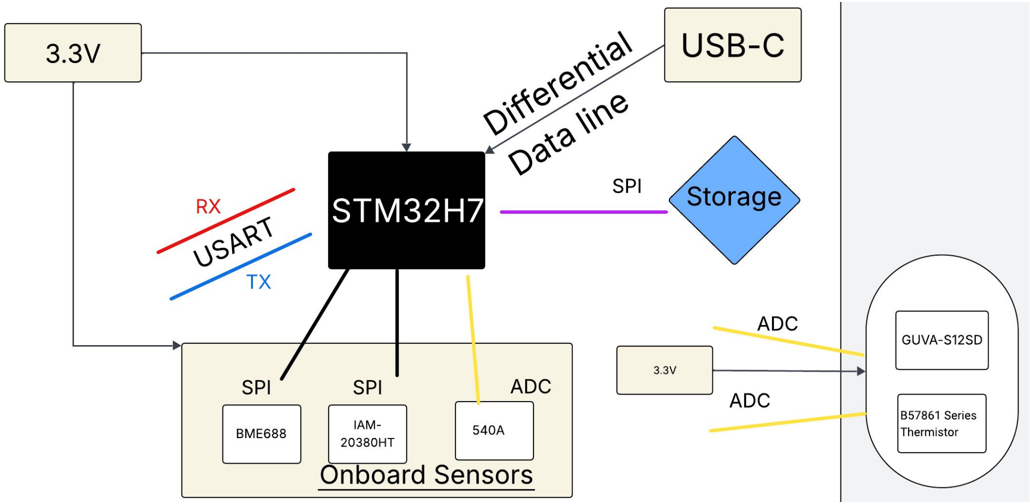




High Power uC subsystem

STM32H7 - High performance Controller

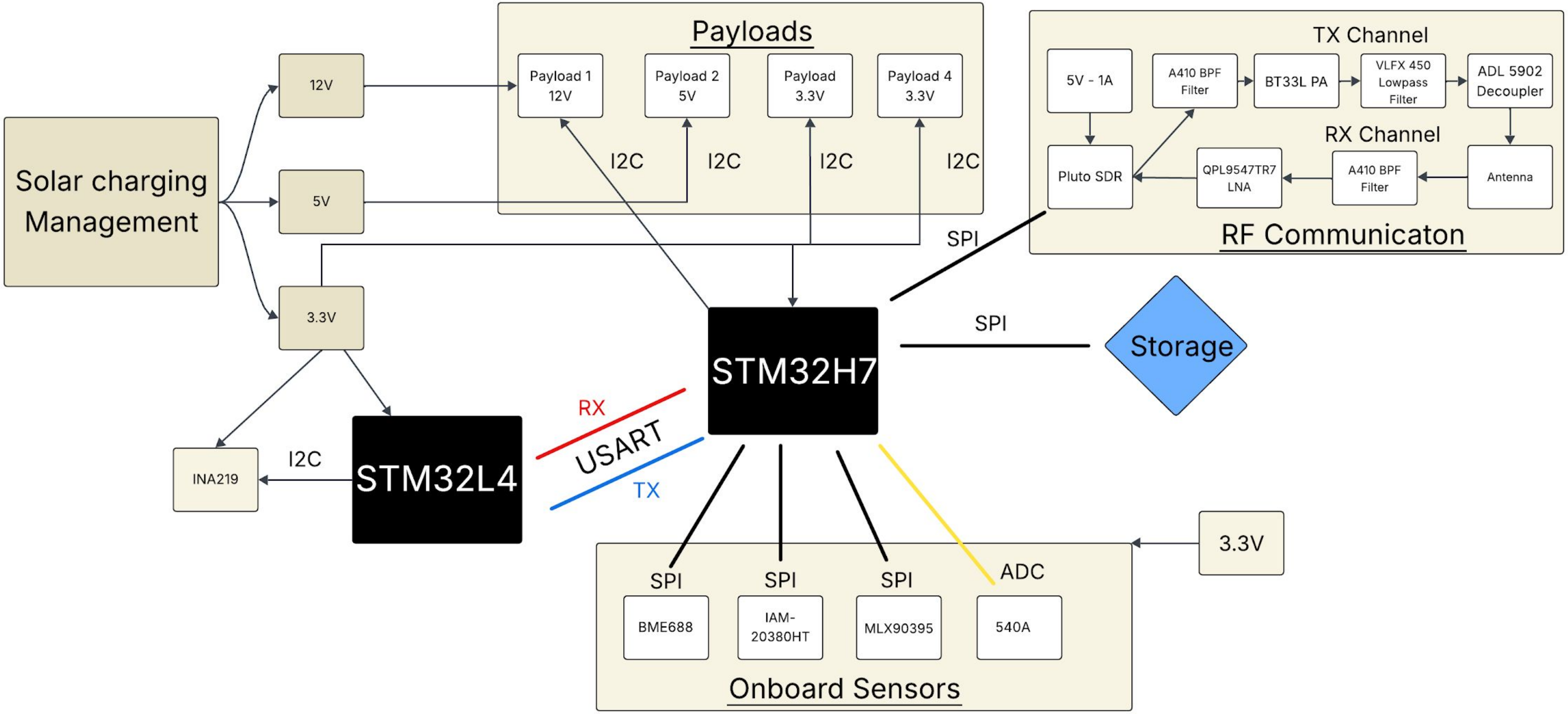
- Powered on only when needed
- Manages onboard sensors and payloads
- Multi-voltage I²C payload interfaces
  - 3.3 V, 5 V, 12 V
- Local data storage (SPI memory)
- RF communication via SPI (PlutoSDR Nano)
- External Sensors



So What ?

High-power sensing, data processing, and communication when active

Final Mainboard Design - Single Board Approach



Future Tests & Risks

Future Tests:

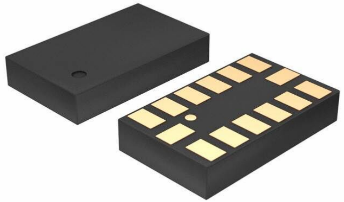
1. L4 and H7 Bring-Up / Programmability
2. H7 ↔ RF Board (PlutoSDR Nano) communication test
3. Verify storage read/write
4. Cold PCB Operation Test (−40°F)

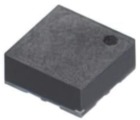
Mainboard Subsystem Risks

1. H7 ↔ RF Board (PlutoSDR Nano) communication fail - **High**
2. Storage communication fails - Med
3. Integrate all subsystems into a single PCB - Med
4. L4 controlling H7 via USART - Low

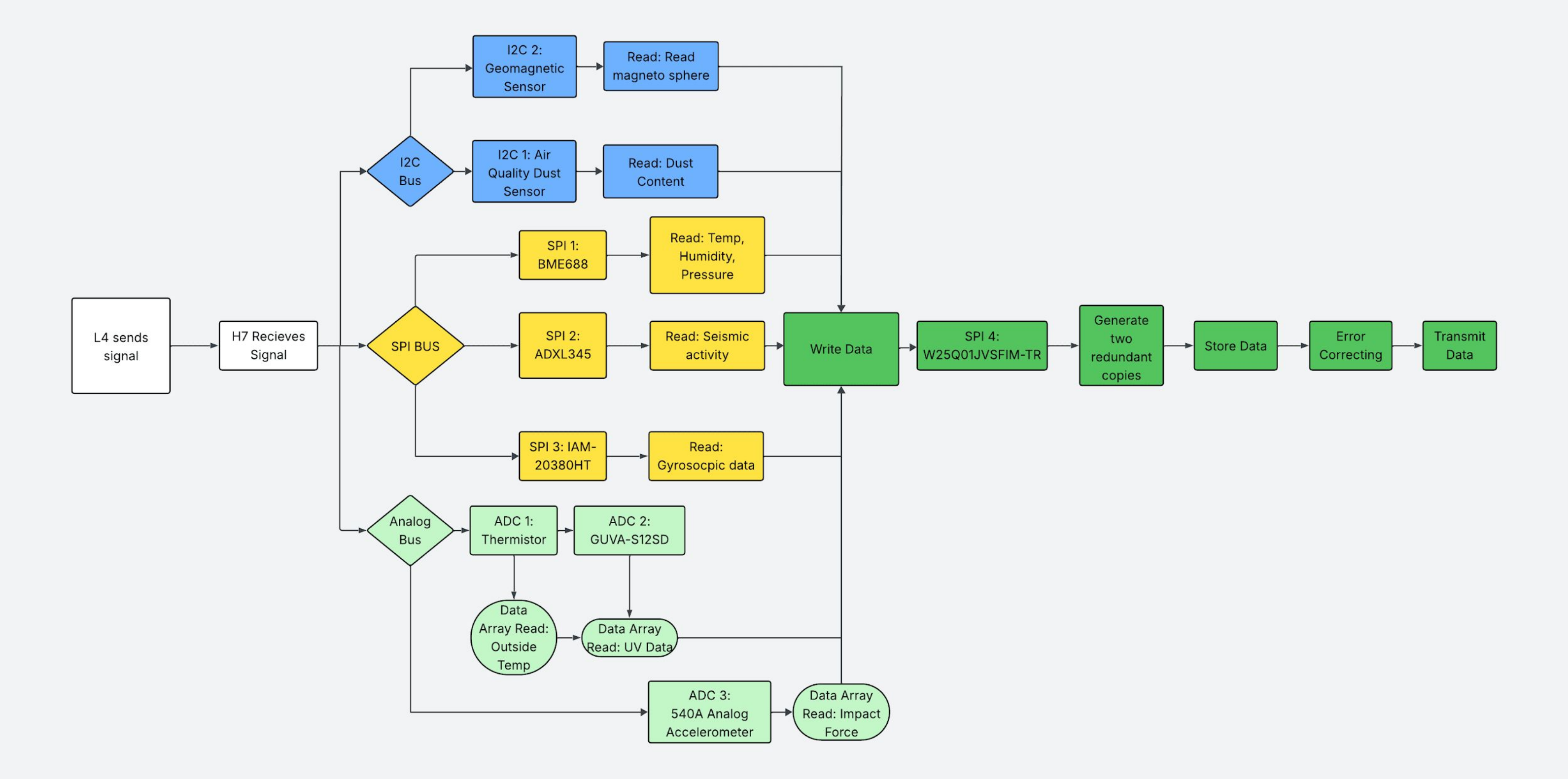
Sensors

- Internal
  - ADXL345 (SPI)
  - BME 688 (SPI)
  - IAM-20380HT (SPI)
  - Snether 540A (Analog)
- External
  - EPCOS NTC Thermistor (Analog)
  - Guava S12-ST (Analog)
  - SPS30 Optical Dust (I2C, Payload)
  - HSCDTD008A (I2C, Payload)





Software Block diagram



Enclosure breakdown

- Blue: Battery
- Gray: Aluminum enclosure
- Brown: Foam

- Simplified handling and installation
- Mainboard sandwiched between polyamide + foam layers for impact absorption

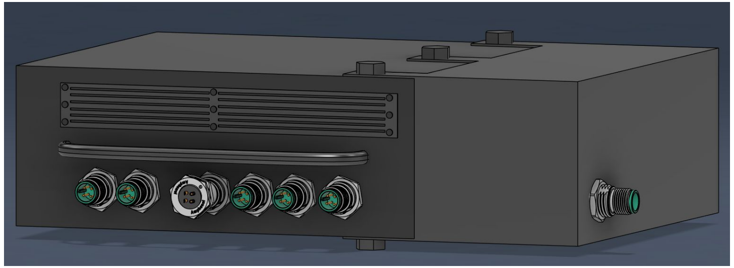


Figure 1: 3D model of enclosure

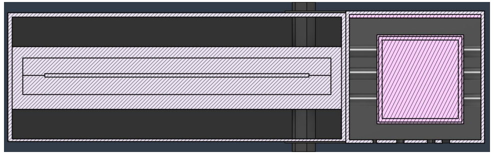


Figure 3: Cross Sectional 2D view of enclosure

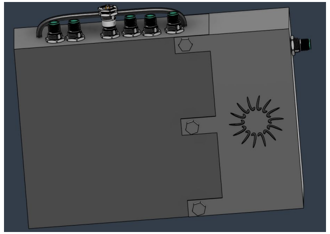


Figure 4: Bottom view of enclosure

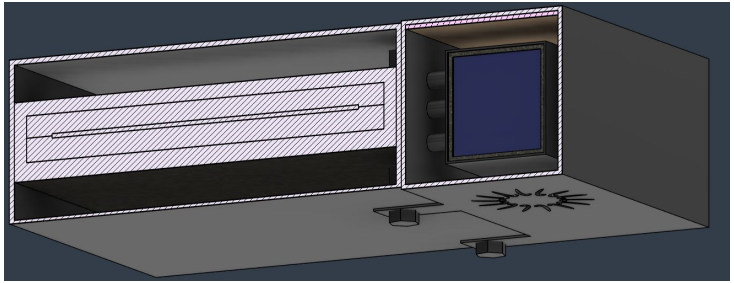


Figure 2: Cross Section view of the enclosure

- Battery isolated in foam, enclosed in aluminum, spring-suspended
- External payload mounting redesigned

Payloads

Design can support 4 payloads:

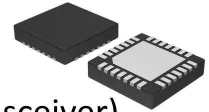
- **2 x Payload Type 1:**
  - V_{MAX}= 3.3v, I_{MAX}= 0.1A
  - Runtime: 10 mins
  - I^2C
- **Payload Type 2:**
  - V_{MAX}= 5v, I_{MAX}= 0.25A
  - Runtime: 10 mins
  - I^2C
- **Payload Type 3:**
  - V_{MAX}= 12v, I_{MAX}= 0.5A
  - Runtime: 10 mins
  - I^2C

Possible Payload:

- **SPS30 Optical Dust sensor:**
  - I^2C / UART
  - Operating voltage: 5 V
  - Current Draw: 55- 60 mA
  - Has a 5-pin female VH connector

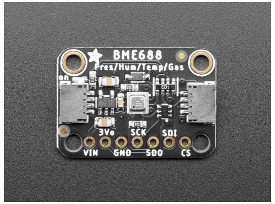
Technology Selection

- **Power:**
  - Samsung 18650
  - 1S Battery Protection Board (COTS)
  - 100W Monocrystalline Solar Panel (Expert Power)

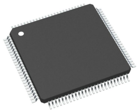


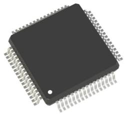
- **Radio:**
  - PlutoSDR Nano (SDR Transceiver)
  - QPL9547TR7 Low Noise Amplifier (Receive)
  - BT33L Amplifier (Transmit)

- **Sensors (In-system):**
  - BME688 (Temp,Humidity,Pressure)
  - IAM-20380HT (Gyro)
  - HSCDTD008A (Magnetometer)
  - 540A (Analog Accelerometer)
  - 1918 (Light Sensor)

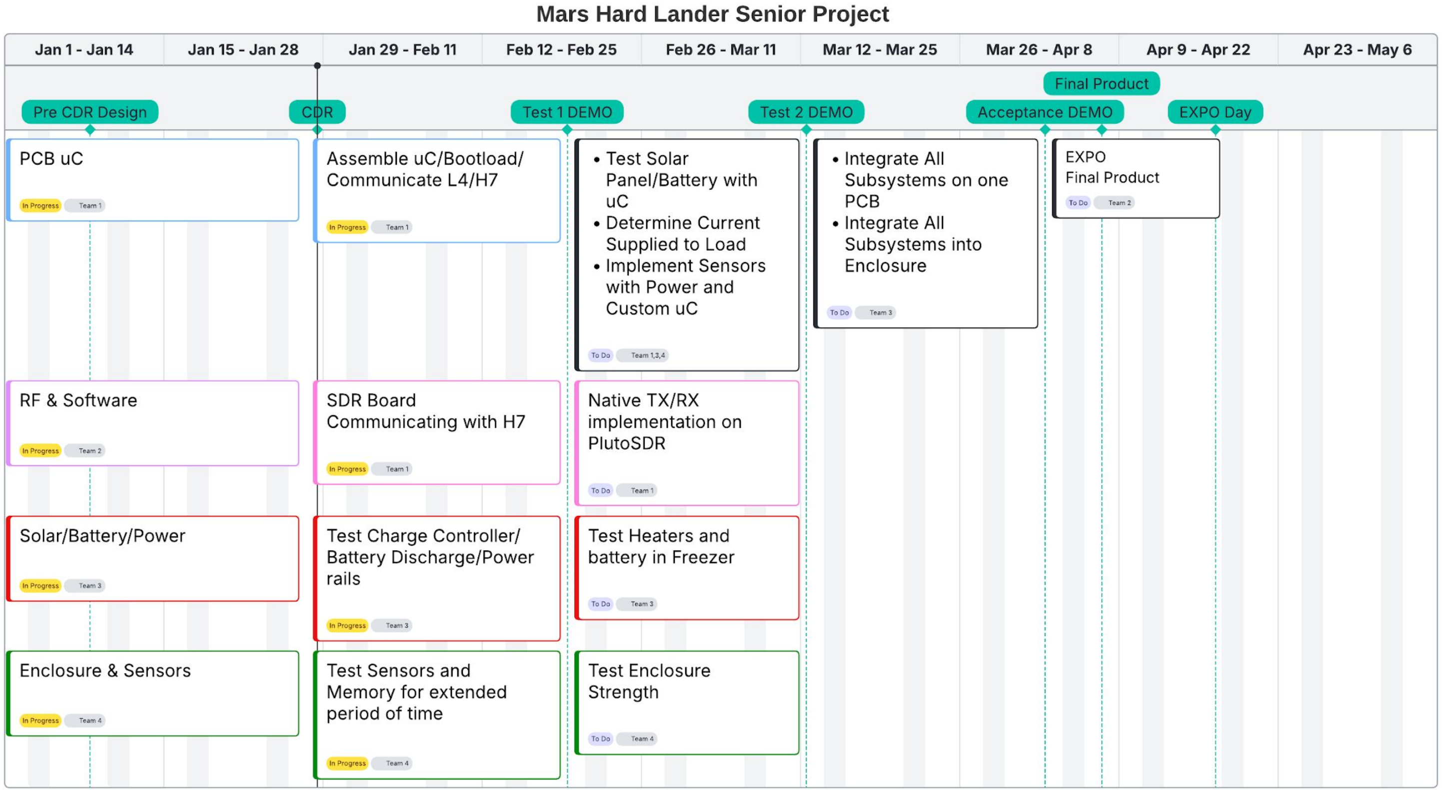


- **Processors and Memory:**
  - STM32H735VHT6
  - STM32L496RGT6
  - W25Q01JVSFIM-TR(Flash Memory)


H7

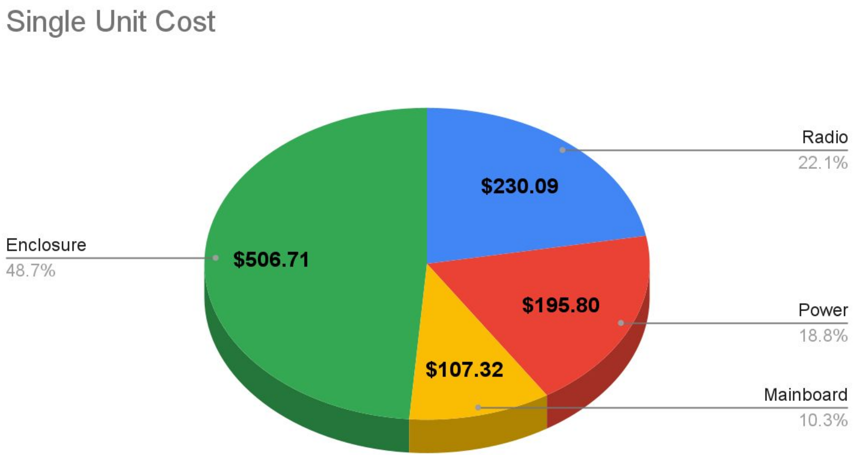

L4

Schedule and Integration Plan (GANTT Chart)

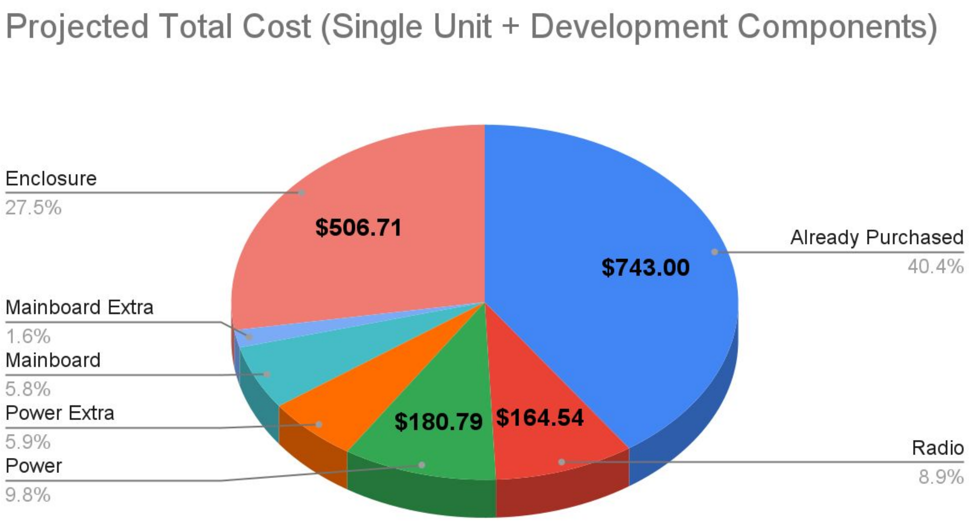


Budgets

Budget



$1,039.92



$1,855.93

Questions?

Schedule and Integration Plan

Integration Test 1 – Core System Bring-Up

- Power system validated (battery + solar + solar charger manager)
- Both STM32L4 & STM32H7 boots and be programmable

Purpose: Prove the system can power up and be controlled

Integration Test 2 – Data & Communication Path

- Sensors → STM32H7 → storage
- STM32H7 ↔ RF board over SPI
- Live data packet transmitted

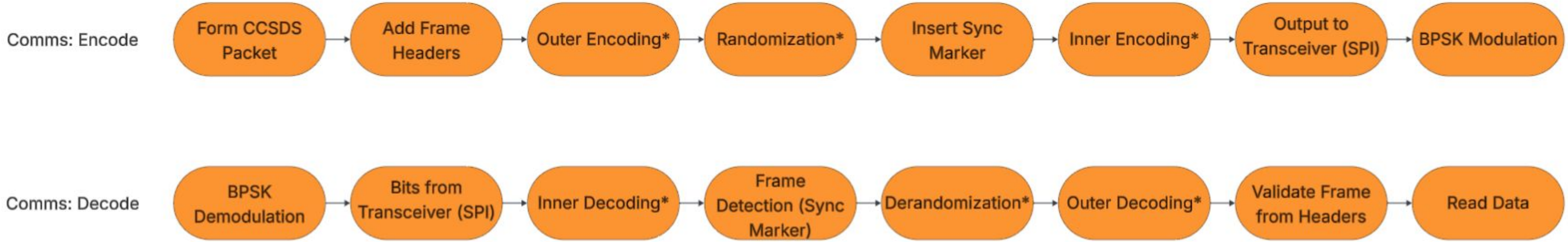
Purpose: Prove data can be collected and transmitted

Acceptance Test – EXPO Readiness

- Single final mainboard
- Battery + solar powered
- L4 & H7 active
- Sensors collecting data
- Storage + RF transmitting
- Enclosure mounted system

Purpose: Demonstrate final system functionality

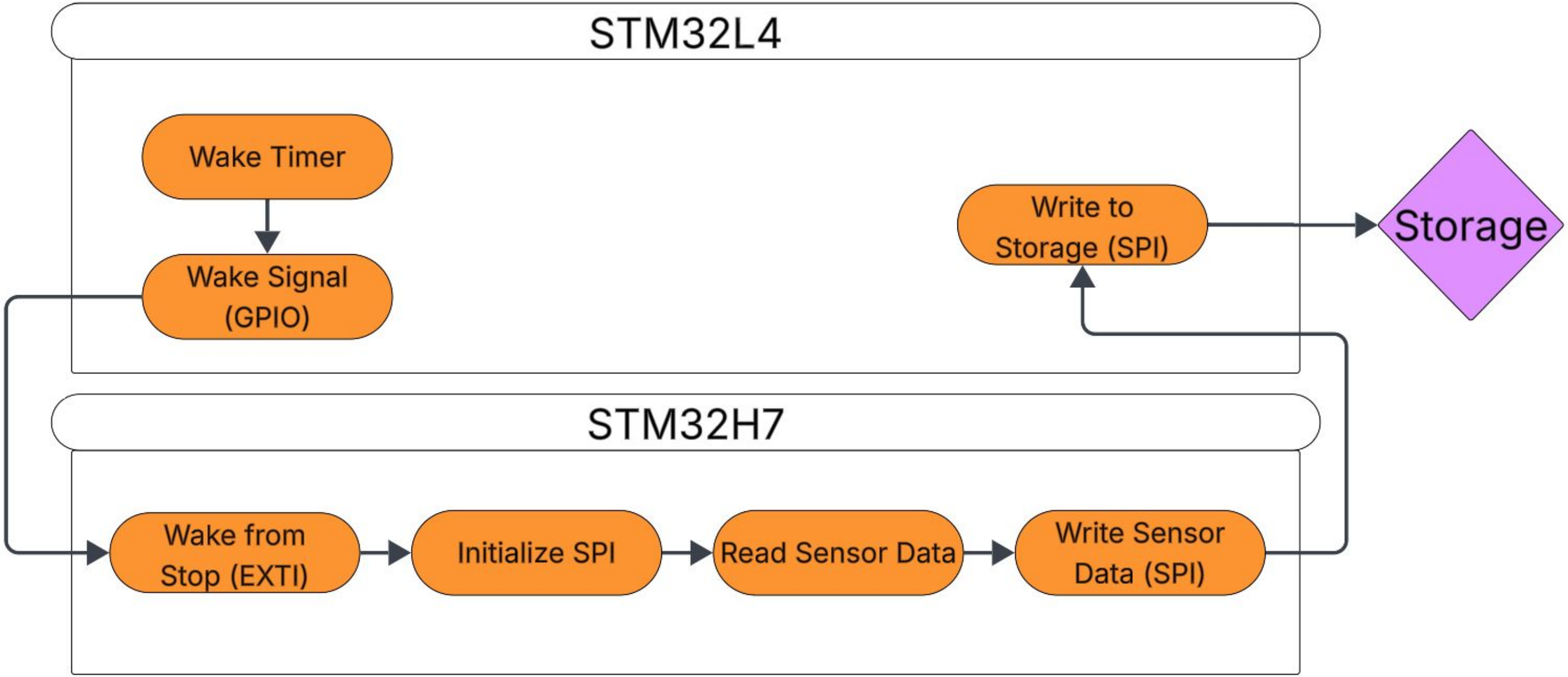
Radio Software Diagram



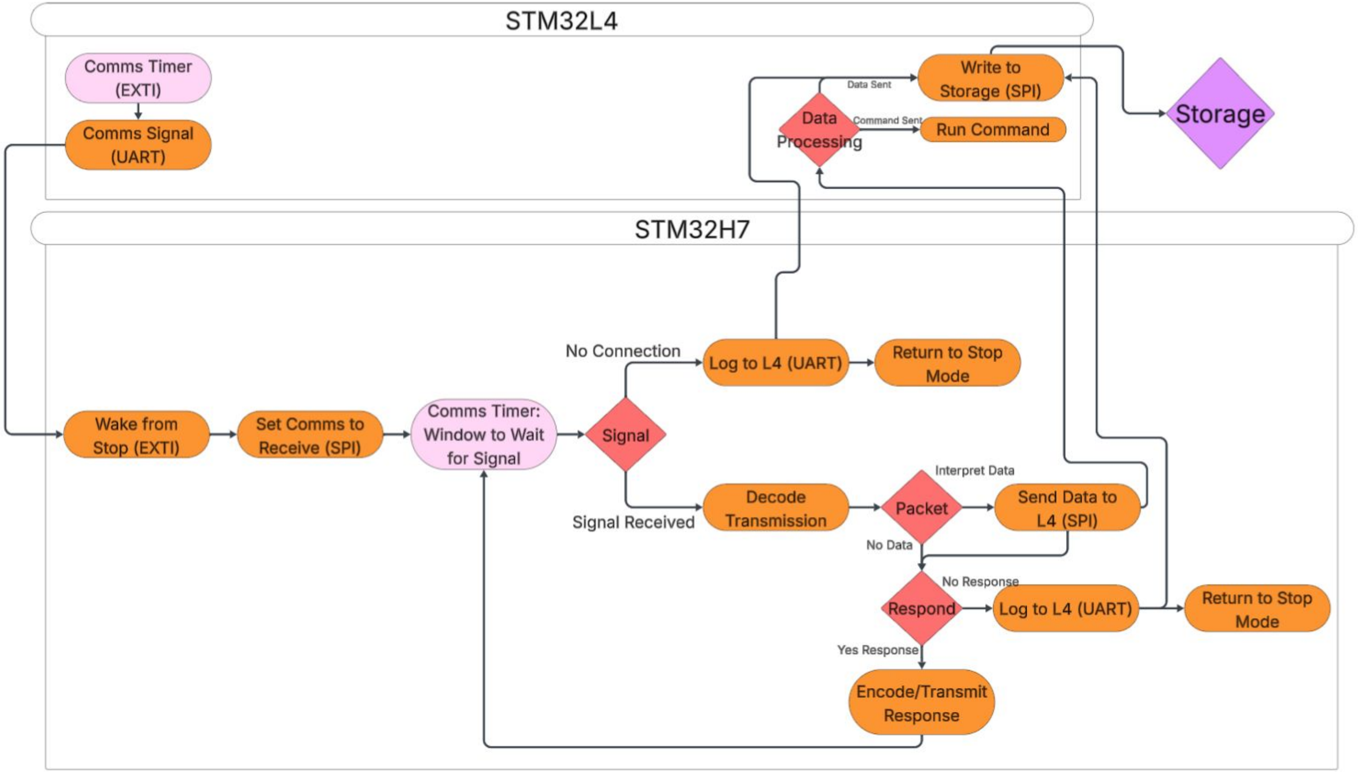
Note:

- Asterisks denote steps that depend on link configuraiton
- In our configuration, Inner encoding is Convolutional, and outer encoding is Reed-Solomon

Software: Sensor Data Collection



Software: Comms Management (outdated)



Power Budget

Without Payload

- Expected Energy usage:
  - 15.5 Wh/day
- Runtime of Battery with 80% of charge:
  - 6 days
- 6 days = 5.84 Sols

With Payloads

- Expected Energy usage:
  - 16.9 Wh/day
- Runtime of Battery with 80% of charge:
  - 5.53 days
- 5.53 days = 5.38 Sols

Hook & Elevator Pitch

Hook:

Traditional soft-landing architectures limit scalability due to mass, cost, and mechanical complexity. A hard-landed platform removes the need for deceleration hardware, enabling a simpler, lighter, and more repeatable architecture for deploying distributed sensor nodes across Mars.

Elevator Pitch:

We are developing a low-cost, high-impact-survivable Mars hard lander that operates as a fixed environmental sensor node. After impact, it powers up autonomously, collects localized environmental data, stores it redundantly, and transmits it to the Mars relay network on a scheduled basis. Because it removes the complexity and cost of soft-landing hardware, this architecture is inherently scalable, repeatable, and deployable as a distributed sensor network for long-duration Martian science.

Design Constraints

- Must integrate with Mars Relay Network for reliable communications.
- Maximum size of 1 m × 1 m footprint to meet strict size limits.
- Engineered for harsh Martian conditions to ensure mission survivability of ~3.7 years.
- Easily reproducible design enabling rapid deployment.
- Cost-optimized architecture for budget-conscious missions.
- Ultra-low-power operation to extend lander lifetime.

Engineering Requirements

- **Environment**
  - ✔ Impact: 200-2000g
  - ✔ Thermal: -45C to 55C
  - ✔ Pressure: 6-7 millibars
  - ✔ Radiation: 76-80 mGy/year (earth year)
  - ✔ Dust/Debris: PCB protection down to 0.5 microns
  - ✔ Heating: maintain 5°F/- 15°C ambient temperature
- **Measurement (Collect Data Every Minute)**
  - ✔ Pressure: 0-1500Pa
  - ✔ Acceleration: 0-3000G
  - ✔ Altitude: -8 to 2 km
  - ✔ Voltage: Two sig. fig. accuracy
  - ✔ Magnetism: +- 2000 nT
- **Network**
  - ✔ Send data every 2 hours within 5 minute window
- **Power**
  - ✔ Power solution must last for 2 mars years
  - ✔ Battery Life must last at least 2 Sol day
- **Functionality**
  - ✔ Overall system must survive for at least 2 Mars years
- **Storage**
  - ✔ At least 1GB onboard data storage
  - ✔ At least 10GB onboard program storage
- **Payload Interfaces**
  - ✔ At least 3 payload interfaces